

Q1. What is Data Loading in Power BI and why is it considered the first step of analysis?

- Data loading in Power BI simply means bringing your data into Power BI from different sources like Excel files, CSV files, databases, or online services. It is the process where raw data is imported so that we can start working on it.

In simple words, without data, there is nothing to analyze. That's why data loading is considered the first step of analysis.

When we load data into Power BI, we can then:

- Clean and transform it (using Power Query)
- Create relationships between tables
- Build reports and dashboards

For example, if you have a COVID dataset or a real estate dataset, you first need to load that dataset into Power BI before doing any calculations or creating visuals.

So, data loading is the foundation of the entire analysis process. Once the data is properly loaded, only then we can move forward with meaningful insights and decision-making.

Q2. Explain the difference between “Load” and “Transform Data” in Power BI.

- In Power BI, when you import data, you usually get two options: Load and Transform Data. Both are used while bringing data into Power BI, but they serve different purposes.

1. Load

The Load option directly imports the data into Power BI without making any changes. It is useful when your data is already clean and properly structured. In simple terms, you are saying: “My data is ready, just bring it in.”

2. Transform Data

The Transform Data option opens the Power Query Editor, where you can clean and modify your data before loading it. This includes tasks like removing duplicates, handling missing values, splitting columns, changing data types, etc. Here you are saying: “Let me fix and prepare the data first.”

- **Difference in simple words**

- Load = Directly use the data (no changes)
- Transform Data = Clean and prepare the data before using it

- **Example**

If your dataset has errors like null values or combined columns (e.g., full name in one column), you should choose Transform Data to fix it first. But if everything is already correct, you can simply click Load.

So, the main difference is that Load is faster but basic, while Transform Data gives you control to clean and shape your data for better analysis.

Q3. What is a Fact Table and a Dimension Table? Give examples from the dataset.

- In Power BI, data is usually organized into two types of tables: Fact Tables and Dimension Tables. These help in making data analysis easier and more structured.

1. Fact Table

A Fact Table contains the main data or measurable values that we want to analyze. These are usually numerical values like totals, counts, or amounts.

In simple words, Fact Table = “What is happening” (data with numbers)

Example (from dataset):

In a COVID dataset, columns like:

- Confirmed Cases
- Recovered Cases
- Deaths
- Active Cases

2. Dimension Table

A Dimension Table contains descriptive information that helps us understand and filter the data in the fact table.

In simple words, Dimension Table = “Details about the data”

Example (from dataset):

Columns like:

- Country
- State
- Date
- Region

Final Understanding

- Fact Table = Numerical data (analysis values)
- Dimension Table = Descriptive data (context of analysis)

For example, if you want to analyze “Total confirmed cases in India on a particular date”

- “Confirmed Cases” comes from the Fact Table,
- while “India” and “Date” come from the Dimension Table.

So, both tables work together to make data analysis in Power BI more meaningful and easy to understand.

Q4. Why is Star Schema preferred over Snowflake Schema in Power BI?

- In Power BI, Star Schema is generally preferred over Snowflake Schema because it is simpler, faster, and easier to understand.

1. What is Star Schema (simple idea)?

In a Star Schema, we have:

- One central Fact Table (main data)
- Connected Dimension Tables (details like country, date, etc.)
- It looks like a star, with one center and multiple connected tables.

2. What is Snowflake Schema?

In a Snowflake Schema:

- Dimension tables are further divided into multiple smaller tables
- This makes the structure more complex (like branches of a snowflake)

3. Why Star Schema is Preferred

I. Easy to Understand

Star Schema has a simple structure, so it is easier for beginners and analysts to work with.

II. Better Performance

Since there are fewer joins between tables, Power BI can process data faster and reports load quickly.

III. Faster Report Building

Creating visuals and dashboards becomes easier because relationships are straightforward.

IV. Less Complexity

Snowflake Schema has too many tables and relationships, which can confuse users and increase chances of errors.

4. Simple Difference

- Star Schema = Simple, fast, user-friendly
- Snowflake Schema = Complex, more tables, slower performance

Q5. Identify and remove duplicate records based on Date, Country, and State.

- In this task, I worked on the dataset named “Corona_Virus_2020” to identify and remove duplicate records. Duplicate data can affect the accuracy of analysis, so it is important to clean it before working further.

First, I loaded the Corona_Virus_2020 dataset into Power BI and opened the Power Query Editor by clicking on “Transform Data”. Then, I selected the columns Date, Country, and State, because duplicates needed to be checked based on these fields.

After selecting these columns, I used the “Remove Duplicates” option in Power Query. This step removed all repeated rows where the combination of Date, Country, and State was the same, keeping only unique records.

As a result, the dataset became clean and more reliable for analysis. This helps in generating accurate reports and insights without any duplication errors.

Q6. Identify and replace null values in Vaccination _Status.

- In this task, I worked on the “Corona_Virus_2020” dataset to identify and handle missing values in the Vaccination _Status column. Null values can create problems in analysis, so it is important to replace them with meaningful values.

First, I loaded the dataset into Power BI and opened the Power Query Editor by clicking on “Transform Data”. Then, I carefully checked the Vaccination _Status column and found that some records had null (blank) values.

To fix this, I selected the Vaccination _Status column and used the “Replace Values” option. I replaced the null values with a suitable value such as “Not Vaccinated” (or any appropriate label based on context). This helped in making the data more clear and usable.

After replacing the null values, the dataset became more complete and easier to analyze. Now, there are no missing entries in the Vaccination _Status column, which improves the accuracy of reports and visualizations.

Q7. Create a new column to calculate Recovery Rate.

- In this task, I worked on the “Corona_Virus_2020” dataset to create a new column that calculates the Recovery Rate. Recovery Rate helps us understand how many people have recovered compared to the total confirmed cases.

First, I loaded the dataset into Power BI and opened the Power Query Editor by clicking on “Transform Data”. After that, I went to the “Add Column” tab and selected the option to create a Custom Column.

Then, I used the formula to calculate Recovery Rate:

$$\text{Recovery Rate} = (\text{Recovered Cases} / \text{Confirmed Cases}) \times 100$$

Using this formula, I created a new column named “Recovery Rate”. This column shows the percentage of recovered cases for each record.

After creating the column, I verified the values to ensure the calculation was correct. This new column helps in better analysis and visualization of how effectively patients are recovering

Q8. Create a summarized table showing total confirmed cases by Country.

- In this task, I worked on the “Corona_Virus_2020” dataset to create a summarized table that shows the total confirmed cases for each country. This helps in understanding how different countries are affected by the virus.

First, I loaded the dataset into Power BI and ensured that the data was properly cleaned. Then, I moved to the Report View (or Data View) to create the summary.

After that, I selected the Country column and the Confirmed Cases column. Using the aggregation feature, I grouped the data by Country and applied the Sum function on Confirmed Cases.

As a result, Power BI created a summarized table where each country shows the total number of confirmed cases.

This summarized table makes it easier to compare countries and analyze which regions have higher or lower case counts.